CHEN ZHIWEI

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RESEARCH INTERESTS

My research enthusiasm lies in tackling the efficiency and reliability problems emerged alongside the development of AI. My recent work focuses on strategic reasoning for LLM agents and efficient multi-agent systems through KV cache reuse.

EDUCATION

National University of Singapore **Aug 2023 - Nov 2024**
Master of Technology in Artificial Intelligence Systems, GPA: 3.51/4.0
Sun Yat-sen University **Sep 2018 - Jun 2022**
Bachelor of Engineering in Traffic Engineering, GPA: 3.5/4.0

RESEARCH EXPERIENCE

The Hong Kong University of Science and Technology (Guangzhou) **Apr 2026 - Present**
Research Assistant | Advisor: Prof. **Tianxiang Zhao**

LatentRemedy: Efficient Remedy For Non-Prefix KV Cache Reuse in Multi-Agent Systems
- Identified conflicting-objective scenarios for non-prefix KV-cache reuse in LLM multi-agent systems, where existing cache-repair baselines fail to recover substantial performance losses.
- Revealed reasoning budget and task information encoded in reused KV states as two key factors governing reuse performance, and demonstrated that repeating the target instruction can mitigate task interference.
- Proposed LatentRemedy, a lightweight target-conditioned adapter that learns continuous remedy tokens to separate source and target task information, improving accuracy while preserving the latency benefits of KV-cache reuse.

Adaptive Strategic Reasoning for LLM Agents in Imperfect-Information Games
- Identified key challenges of strategic reasoning, opponent adaptation, and effective use of historical feedback for adaptive decision-making of LLM agents in repeated imperfect-information interactions.
- Proposed SAGE, a training-free strategic reasoning framework that integrates equilibrium-based policies, opponent belief modeling, and memory-based guidance without updating the underlying LLM parameters.
- Evaluated SAGE on Leduc Hold'em, Liar's Dice, and Goofspiel against six game-specific opponent types in each environment, outperforming existing LLM-based game-playing agents.

Center for Artificial Intelligence and Robotics, Chinese Academy of Sciences, Hong Kong
Intern → Research Assistant | Advisor: Prof. **Mingyang Zhao** **Mar 2024 - Feb 2026**

- Built data acquisition and PDF-processing pipelines for multimodal dataset construction and retrieval-augmented generation (RAG).
- Evaluated multimodal models on medical visual question answering and visual reasoning tasks.
- Conducted multimodal representation learning research on ultrasound image captioning and image-text alignment.

PUBLICATIONS

- Z. Rao, P. Zhou, D. Chong, **Z. Chen**, S. Wang, N. Tang. "Structural and Disentangled Adaptation of Large Vision Language Models for Multimodal Recommendation". *SIGIR, 2026*
- Z. Chen, K. Zhou, Q. Lei, **Z. Chen**, K. Cao, L. Qin, X. Yuan, S. Chang, H. Liu, G. Meng. "Joint Thyroid and Nodule Segmentation with Unpaired Labels: From Dual-Branch Learning to Semantic Query Unification". *BIBM, 2026*

MANUSCRIPTS

- **Z. Chen**, T. Wang, Z. Rao, H. Zhu, D. Cao, T. Zhao. “SAGE: Structured Strategic Reasoning for Efficient LLM Game Playing”. preprint, *2026*